C.A.N.I.S.
Corrosion And Nuclear Inspection System

Drum Store Inspection Report

Warehouse v3, aisle A (simulated) | 2026-11-08

Store	Warehouse v3, aisle A (simulated)	Report number	CANIS-canis_aisle_patrol
Inspected on	2026-11-08 12:00:36 UTC	Time taken	4 min 38 s
Inspected by	C.A.N.I.S. inspection robot	Status	Practice run in a simulated store

Summary

Drums checked	**36 of 36**	Areas with low gamma	**8**
• Need action	**6**	Areas restricted	**0**
• Check again	**0**	Areas to keep out of	**0**
• No problem found	**30**	Areas not checked	**1**
Radiation dose to the robot	**0% of its limit**	Time taken	**4 min 38 s**

6 drums need action. 1 area was not checked.

How to read this report

- **Action needed.** Damage seen clearly. An engineer should assess the drum.
- **Check again.** Possible damage, not yet confirmed.
- **No problem found** in this inspection.

Low gamma Gamma below the demo threshold.

Restricted Simulation flags this section for review.

Keep out Highest priority simulation flag.

Not checked The robot did not measure here: treat as unknown.

Radiation is given in microsieverts per hour (µSv/h). Simulation flags start above 7.5 and escalate above 100 µSv/h.

1 Aisle survey results

The robot measured radiation as it walked. Each aisle is split into 2-metre sections, and each section is given a simulation rating. Site staff must assess access separately.

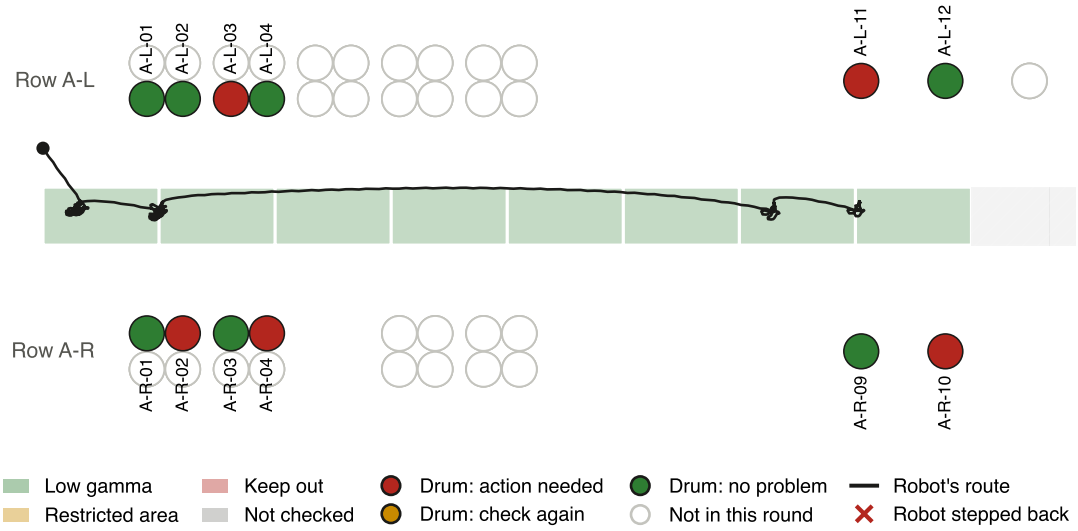


Figure 1. Plan of the store. Coloured bands are the aisle sections; circles are drums (stacked drums share one circle, coloured by the most urgent).

Where	Section	Rating	Why
Aisle between rows A-R and A-L	0–2 m	Low gamma	Highest logged gamma was 2.0 µSv/h.
Aisle between rows A-R and A-L	2–4 m	Low gamma	Highest logged gamma was 2.5 µSv/h.
Aisle between rows A-R and A-L	4–6 m	Low gamma	Highest logged gamma was 2.8 µSv/h.
Aisle between rows A-R and A-L	6–8 m	Low gamma	Highest logged gamma was 2.6 µSv/h.
Aisle between rows A-R and A-L	8–10 m	Low gamma	Highest logged gamma was 2.1 µSv/h.
Aisle between rows A-R and A-L	10–12 m	Low gamma	Highest logged gamma was 1.8 µSv/h.
Aisle between rows A-R and A-L	12–14 m	Low gamma	Highest logged gamma was 1.9 µSv/h.
Aisle between rows A-R and A-L	14–16 m	Low gamma	Highest logged gamma was 2.0 µSv/h.
Aisle between rows A-R and A-L	16–18 m	Not checked	The robot did not go here this time.

Sections are measured along the aisle from the end nearest drum 1. Hydrogen stayed below the sensor's 2 ppm detection limit throughout.

2 Every drum at a glance

All drums in this inspection, most urgent first.

Drum	Waste	Where	Result	What was seen	How sure	Radiation
A-L-03-T1	Low-level	Left rack, bay 3, shelf 1	Action needed	Bulging	High (97%)	Normal (1.6 µSv/h)
A-L-03-T3	Low-level	Left rack, bay 3, shelf 3	Action needed	Rust	Medium (65%)	Normal (1.8 µSv/h)
A-L-11-T2	Intermediate-level	Left rack, bay 11, shelf 2	Action needed	Rust	Medium (66%)	Normal (1.4 µSv/h)
A-R-02-T2	Low-level	Right rack, bay 2, shelf 2	Action needed	Rust	High (96%)	Normal (1.2 µSv/h)
A-R-04-T2	Low-level	Right rack, bay 4, shelf 2	Action needed	Rust	High (97%)	Normal (1.8 µSv/h)
A-R-10-T2	Intermediate-level	Right rack, bay 10, shelf 2	Action needed	Rust	Medium (52%)	Normal (1.6 µSv/h)
A-L-01-T1	Low-level	Left rack, bay 1, shelf 1	No problem found	Nothing found	High (97%)	Normal (1.1 µSv/h)
A-L-01-T2	Low-level	Left rack, bay 1, shelf 2	No problem found	Nothing found	High (96%)	Normal (1.2 µSv/h)
A-L-01-T3	Low-level	Left rack, bay 1, shelf 3	No problem found	Nothing found	High (95%)	Normal (1.1 µSv/h)
A-L-02-T1	Low-level	Left rack, bay 2, shelf 1	No problem found	Nothing found	High (96%)	Normal (1.2 µSv/h)
A-L-02-T2	Low-level	Left rack, bay 2, shelf 2	No problem found	Nothing found	High (96%)	Normal (1.1 µSv/h)
A-L-02-T3	Low-level	Left rack, bay 2, shelf 3	No problem found	Nothing found	High (95%)	Normal (1.2 µSv/h)
A-L-03-T2	Low-level	Left rack, bay 3, shelf 2	No problem found	Nothing found	High (96%)	Normal (1.8 µSv/h)
A-L-04-T1	Low-level	Left rack, bay 4, shelf 1	No problem found	Nothing found	High (96%)	Normal (1.6 µSv/h)
A-L-04-T2	Low-level	Left rack, bay 4, shelf 2	No problem found	Nothing found	High (96%)	Normal (1.6 µSv/h)
A-L-04-T3	Low-level	Left rack, bay 4, shelf 3	No problem found	Nothing found	High (96%)	Normal (1.7 µSv/h)
A-L-11-T1	Intermediate-level	Left rack, bay 11, shelf 1	No problem found	Nothing found	High (94%)	Normal (1.2 µSv/h)
A-L-11-T3	Intermediate-level	Left rack, bay 11, shelf 3	No problem found	Nothing found	High (95%)	Normal (1.4 µSv/h)
A-L-12-T1	Intermediate-level	Left rack, bay 12, shelf 1	No problem found	Nothing found	High (92%)	Normal (1.5 µSv/h)
A-L-12-T2	Intermediate-level	Left rack, bay 12, shelf 2	No problem found	Nothing found	High (86%)	Normal (1.4 µSv/h)
A-L-12-T3	Intermediate-level	Left rack, bay 12, shelf 3	No problem found	Nothing found	High (94%)	Normal (1.5 µSv/h)
A-R-01-T1	Low-level	Right rack, bay 1, shelf 1	No problem found	Nothing found	High (96%)	Normal (1.2 µSv/h)
A-R-01-T2	Low-level	Right rack, bay 1, shelf 2	No problem found	Nothing found	High (96%)	Normal (1.2 µSv/h)
A-R-01-T3	Low-level	Right rack, bay 1, shelf 3	No problem found	Nothing found	High (96%)	Normal (1.2 µSv/h)
A-R-02-T1	Low-level	Right rack, bay 2, shelf 1	No problem found	Nothing found	High (96%)	Normal (1.2 µSv/h)
A-R-02-T3	Low-level	Right rack, bay 2, shelf 3	No problem found	Nothing found	High (93%)	Normal (1.2 µSv/h)
A-R-03-T1	Low-level	Right rack, bay 3, shelf 1	No problem found	Nothing found	High (95%)	Normal (1.9 µSv/h)
A-R-03-T2	Low-level	Right rack, bay 3, shelf 2	No problem found	Nothing found	High (96%)	Normal (1.8 µSv/h)
A-R-03-T3	Low-level	Right rack, bay 3, shelf 3	No problem found	Nothing found	High (95%)	Normal (1.6 µSv/h)
A-R-04-T1	Low-level	Right rack, bay 4, shelf 1	No problem found	Nothing found	High (96%)	Normal (1.8 µSv/h)
A-R-04-T3	Low-level	Right rack, bay 4, shelf 3	No problem found	Nothing found	High (95%)	Normal (1.8 µSv/h)
A-R-09-T1	Intermediate-level	Right rack, bay 9, shelf 1	No problem found	Nothing found	High (96%)	Normal (1.4 µSv/h)
A-R-09-T2	Intermediate-level	Right rack, bay 9, shelf 2	No problem found	Nothing found	High (95%)	Normal (1.5 µSv/h)
A-R-09-T3	Intermediate-level	Right rack, bay 9, shelf 3	No problem found	Nothing found	High (95%)	Normal (1.4 µSv/h)
A-R-10-T1	Intermediate-level	Right rack, bay 10, shelf 1	No problem found	Nothing found	High (96%)	Normal (1.5 µSv/h)
A-R-10-T3	Intermediate-level	Right rack, bay 10, shelf 3	No problem found	Nothing found	High (94%)	Normal (1.6 µSv/h)

"How sure" is how confident the camera software was about the damage it saw. This is the first inspection of these drums, so there is nothing to compare with yet. Not measured in this inspection: surface wipe (swab) results.

3 Drums that need attention

Drum A-L-03-T1

Action needed

Low-level waste | Left rack, bay 3, shelf 1



Photo taken 1:32 into the inspection. The bold outline marks drum A-L-03-T1.

What was seen	Bulging
How sure	High (97%)
Why this rating	Bulging recorded as a confirmed finding.
Radiation at drum	Normal (1.6 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	+0.6 °C compared with the air

What to do: Limit access to this aisle. Ask an engineer to assess the drum. Inspect again at the next round.

Drum A-L-03-T3

Action needed

Low-level waste | Left rack, bay 3, shelf 3



Photo taken 2:24 into the inspection. The bold outline marks drum A-L-03-T3.

What was seen	Rust
How sure	Medium (65%)
Why this rating	Rust recorded as a confirmed finding.
Radiation at drum	Normal (1.8 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	-0.1 °C compared with the air

What to do: Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

Drum A-L-11-T2

Action needed

Intermediate-level waste | Left rack, bay 11, shelf 2

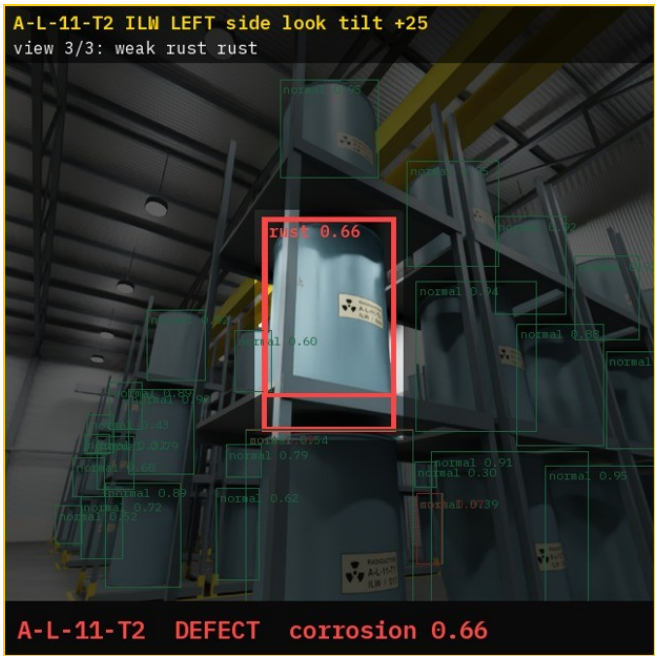


Photo taken 3:19 into the inspection. The bold outline marks drum A-L-11-T2.

What was seen	Rust
How sure	Medium (66%)
Why this rating	Rust recorded as a confirmed finding.
Radiation at drum	Normal (1.4 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	+1.3 °C compared with the air

What to do: Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

Drum A-R-02-T2

Action needed

Low-level waste | Right rack, bay 2, shelf 2

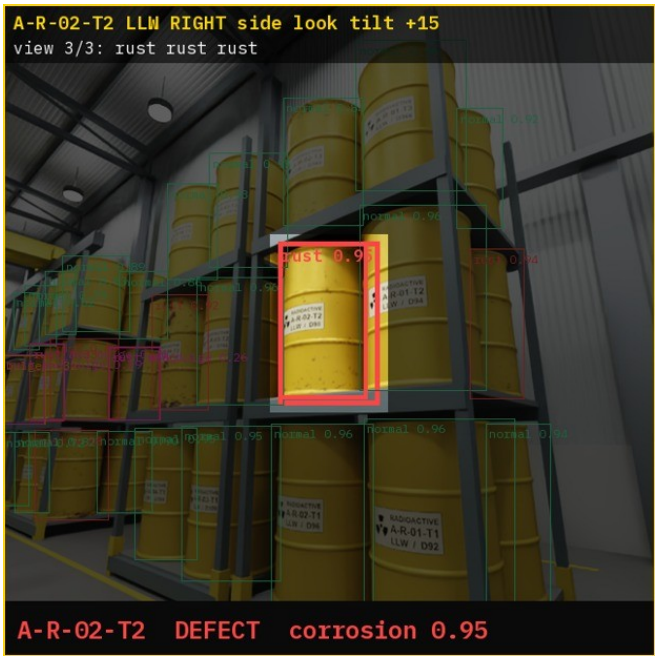


Photo taken 0:49 into the inspection. The bold outline marks drum A-R-02-T2.

What was seen	Rust
How sure	High (96%)
Why this rating	Rust recorded as a confirmed finding.
Radiation at drum	Normal (1.2 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	+0.2 °C compared with the air

What to do: Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

Drum A-R-04-T2

Action needed

Low-level waste | Right rack, bay 4, shelf 2



Photo taken 2:15 into the inspection. The bold outline marks drum A-R-04-T2.

What was seen	Rust
How sure	High (97%)
Why this rating	Rust recorded as a confirmed finding.
Radiation at drum	Normal (1.8 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	-0.1 °C compared with the air

What to do: Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

Drum A-R-10-T2

Action needed

Intermediate-level waste | Right rack, bay 10, shelf 2

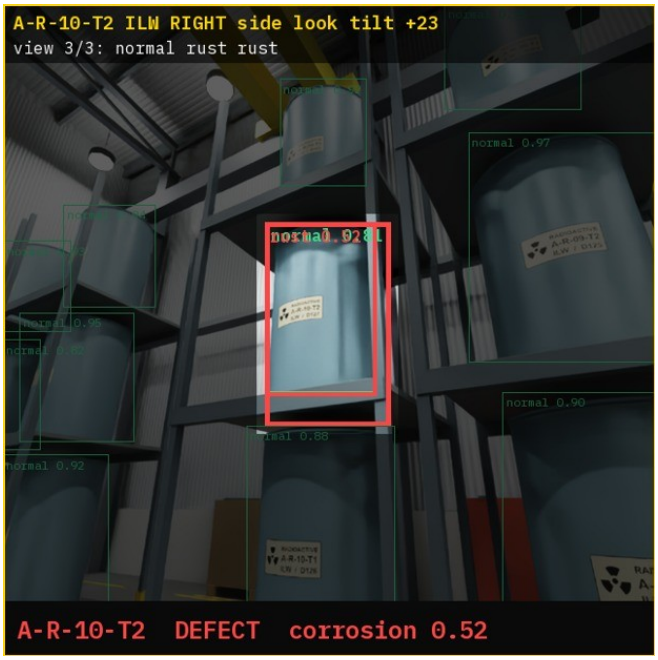


Photo taken 4:16 into the inspection. The bold outline marks drum A-R-10-T2.

What was seen	Rust
How sure	Medium (52%)
Why this rating	Rust recorded as a confirmed finding.
Radiation at drum	Normal (1.6 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	+1.1 °C compared with the air

What to do: Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

4 What was not checked

- Aisle between rows A-R and A-L, sections 16–18 m: the robot did not walk here this time, so these are unknown.

Not measured anywhere in this inspection: surface wipe (swab) results.

5 How accurate was the robot?

This was a practice run in a simulated store, so we know which drums were really damaged. The robot was not told; the comparison is made only for this report.

- It found 5 of the 9 damaged drums.
- It missed A-L-12-T1, A-R-02-T3, A-R-09-T1, A-R-09-T2.
- It also raised 1 false alarm: A-R-10-T2 was actually undamaged.

A false alarm costs a person an extra check. A missed drum is the more serious error.

6 How this report was made

The robot walks the aisles with a camera and a radiation meter. Software trained to recognise rust and bulging (swelling) on drums points out possible damage in the camera images. When it is unsure, the robot goes back for a closer look from another angle. Fixed safety rules, the same every time, then decide each rating in this report. An AI assistant may add a short written explanation for drums that need action, but it cannot change any rating or number. Everything in this report comes from what the robot recorded during the inspection.

Limits used

What	Level	Source
Restricted area	above 7.5 µSv/h	HSE, *Work with ionising radiation* (L121), regulation 17. Used here as an on-the-spot reading.
Keep out	above 100 µSv/h	HSE, L121 guidance on high instantaneous dose rates.
Surface wipe	above 0.4 / 4 Bq/cm^2	IAEA, *Regulations for the Safe Transport of Radioactive Material* (SSR-6), para. 508.
Warm drum	5 °C above the air	Our own rule for this project; not an official limit.
Hydrogen gas	no level set yet	We have not yet found an official alarm level to cite, so hydrogen does not change any rating.

Please note. This inspection took place in a computer-simulated store, not a real one. Radiation values come from a simplified model. The damage-spotting software was trained on computer-generated images. How far rust has spread is estimated from photos, not measured. Only a surface wipe can show contamination. The format follows the themes of the Nuclear Decommissioning Authority's *Guidance on Monitoring of Waste Packages During Storage* (WPS/640).

A Radiation during the inspection

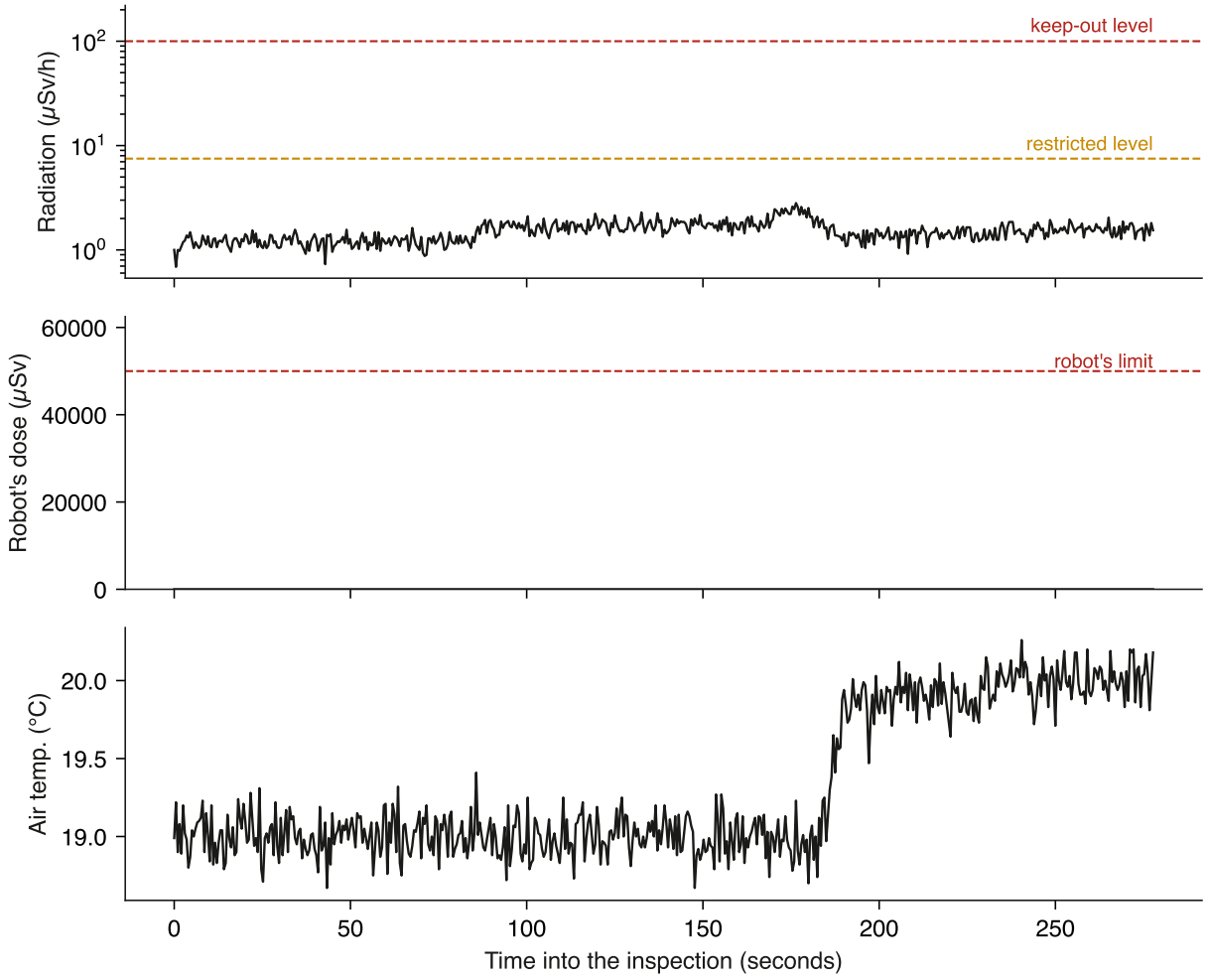


Figure 2. Top: radiation measured by the robot as it walked (dashed lines are the restricted and keep-out levels). Bottom: the total dose the robot received, against its limit. The shaded band is the radiation event.

B Step-by-step record

What the robot did and why, in order. Times are minutes:seconds into the inspection.

Time	Drum	What happened
0:09	A-L-01-T1	No damage seen in 3 camera views.
0:13	A-L-01-T2	No damage seen in 3 camera views.
0:16	A-L-01-T3	Top-tier drum: moved for a closer second look before deciding.
0:22	A-L-02-T1	No damage seen in 2 camera views.
0:26	A-L-02-T2	No damage seen in 3 camera views.
0:29	A-L-02-T3	Top-tier drum: moved for a closer second look before deciding.
0:35	A-R-01-T1	No damage seen in 3 camera views.
0:39	A-R-01-T2	No damage seen in 3 camera views.
0:42	A-R-01-T3	Top-tier drum: moved for a closer second look before deciding.
0:45	A-R-02-T1	No damage seen in 3 camera views.
0:49	A-R-02-T2	Rust seen in 3 of 3 camera views. Recorded as damage.

Time	Drum	What happened
0:53	A-R-02-T3	Top-tier drum: moved for a closer second look before deciding.
0:59	A-L-01-T3	No damage seen in 8 camera views.
1:06	A-R-01-T3	No damage seen in 8 camera views.
1:14	A-L-02-T3	No damage seen in 8 camera views.
1:21	A-R-02-T3	No damage seen in 8 camera views.
1:32	A-L-03-T1	Bulging seen in 3 of 3 camera views. Recorded as damage.
1:35	A-L-03-T2	No damage seen in 3 camera views.
1:39	A-L-03-T3	Top-tier drum: moved for a closer second look before deciding.
1:44	A-L-04-T1	No damage seen in 2 camera views.
1:50	A-L-04-T2	No damage seen in 3 camera views.
1:55	A-L-04-T3	Top-tier drum: moved for a closer second look before deciding.
2:00	A-R-03-T1	No damage seen in 3 camera views.
2:04	A-R-03-T2	No damage seen in 3 camera views.
2:08	A-R-03-T3	Top-tier drum: moved for a closer second look before deciding.
2:11	A-R-04-T1	No damage seen in 3 camera views.
2:14	A-R-04-T2	Rust seen in 3 of 3 camera views. Recorded as damage.
2:18	A-R-04-T3	Top-tier drum: moved for a closer second look before deciding.
2:24	A-L-03-T3	Rust seen in 3 of 8 camera views. Recorded as damage.
2:32	A-R-03-T3	No damage seen in 8 camera views.
2:39	A-L-04-T3	No damage seen in 8 camera views.
2:45	A-R-04-T3	No damage seen in 8 camera views.
3:15	A-L-11-T1	No damage seen in 3 camera views.
3:19	A-L-11-T2	Rust seen in 2 of 3 camera views. Recorded as damage.
3:23	A-L-11-T3	Top-tier drum: moved for a closer second look before deciding.
3:28	A-R-09-T1	No damage seen in 3 camera views.
3:32	A-R-09-T2	No damage seen in 3 camera views.
3:36	A-R-09-T3	Top-tier drum: moved for a closer second look before deciding.
3:42	A-L-11-T3	No damage seen in 8 camera views.
3:48	A-R-09-T3	No damage seen in 8 camera views.
3:59	A-L-12-T1	No damage seen in 3 camera views.
4:03	A-L-12-T2	No damage seen in 3 camera views.
4:06	A-L-12-T3	Top-tier drum: moved for a closer second look before deciding.
4:12	A-R-10-T1	No damage seen in 3 camera views.
4:16	A-R-10-T2	Rust seen in 2 of 3 camera views. Recorded as damage.
4:19	A-R-10-T3	Top-tier drum: moved for a closer second look before deciding.
4:26	A-L-12-T3	No damage seen in 8 camera views.
4:34	A-R-10-T3	No damage seen in 8 camera views.

C Technical details

For engineers and auditors.

Run ID	`canis_aisle_patrol`
Software version	`22157ad`
Damage detector run	YOLO26s whole-barrel conditions (expanded, 8 Nov 2026)
Explanation model	not used in this round
Simulator	NVIDIA Isaac Sim
Status names	Action needed = `Confirmed defect`; Check again = `Suspect`; No problem found = `OK`; Not inspected = `Couldn't inspect`